

JIQI GU

PhD Researcher in Ecology | Community Ecology · Functional Traits · Eco-evolutionary Modelling
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RESEARCH PROFILE

I study how environmental filtering, functional traits, dispersal and evolutionary history shape biodiversity across space and time. My work combines field ecology, global databases and quantitative models, from bryophyte community assembly to historical biogeography and multi-taxon community dynamics.

RESEARCH CONTRIBUTIONS

Trait-based biodiversity and community assembly

I am developing WorldBryDat, a global bryophyte functional-trait database covering 6,123 species and approximately 264 traits (195 morphological traits, 12 reproductive traits, 49 life-history traits, 8 habitat preference traits) [31]. It provides a basis for comparing bryophyte ecological strategies across environments and lineages.

My research on Tibetan bryophytes combines field surveys (551 plots, 1,008 species, 264 traits), functional traits, phylogeny and HMSC to examine environmental filtering and community assembly [28]. Related work investigates bryophyte distributions, habitat associations and intraspecific trait variation [19, 16, 15, 12, 10, 9, 7], bryophyte taxonomy and records [23, 14, 11], and plant community diversity and phylogenetic structure [21, 18, 13].

Eco-evolutionary modelling and historical biogeography

I am developing HmscEE, a spatially explicit framework that extends joint species distribution modelling to study biodiversity through deep time [30]. It combines species occurrences, functional traits, phylogenies, fossil evidence, palaeoenvironmental reconstructions, continental movement and dispersal opportunities. Its bryophyte case study uses approximately 4.4 million occurrence records across 515 million years.

My methodological work includes joint species distribution models and the HMSC framework [24], contributions to JSDM Studio software [29], and research on large language models in ecology [17].

Global biodiversity dynamics and species distribution modelling

Using LIFEPLAN observations, I examine how urbanization relates to community turnover, recurrence and temporal stability across 3,426 bird species and 462,956 molecular taxa of arthropods and fungi [27].

I have contributed to studies of marine fish species distributions using expert maps and occurrence records [26, 22], and to evaluations of species distribution model performance and cross-validation [25, 20].

Earlier empirical and applied ecology

My earlier research spans bryophyte ecology and neural-network methods [6], soil nitrogen responses to freeze–thaw cycles [5], and water-quality assessment, ecological risk monitoring and related applications of artificial intelligence [8, 4, 3, 2, 1].

EDUCATION AND RESEARCH POSITIONS

2025–present | University of Jyväskylä, Finland
Joint PhD Researcher; Supervisor: Prof. Otso Ovaskainen. China Scholarship Council fellow (202506040049).

2023–present | Beijing Normal University, China
PhD Candidate in Ecology; Supervisor: Prof. Anru Lou.

2020–2023 | China Agricultural University, China
MSc, Resource Utilization and Plant Protection; Supervisor: Prof. Xiaoming Shao.

2016–2020 | Northeast Agricultural University, China
BSc, Ecology.

METHODS AND RESEARCH SKILLS

HMSC/JSDMs, SDMs, GEN3SIS, SEM, mixed-effects models; functional-trait, phylogenetic and community-stability analyses; multivariate statistics (CCA, RDA, NMDS); R, GIS, Google Earth Engine, machine learning; bryophyte identification, ecological fieldwork and trait measurements.

TEACHING AND KNOWLEDGE EXCHANGE

Advanced Experimental Design — Instructor, PhD course, Northeast Agricultural University (2023). Taught field sampling, experimental design, ordination, structural equation models, SDMs/JSDMs and mixed-effects models.

Joint Species Distribution Modelling — Instructor, Chinese Academy of Sciences online public courses (2024 and 2026; more than 7,000 participants per course). I have also shared tutorials on HMSC, GEN3SIS and community ecology (approximately 96,000 cumulative reads).

Large Language Models in Ecological Research — Invited lectures (more than 8,000 participants; approximately 18,000 livestream views), alongside a related publication [17].

ACADEMIC SERVICE, OUTREACH AND HONOURS

Peer review: *Conservation Biology*; *Methods in Ecology and Evolution*; *Global Ecology and Biogeography*.

Outreach: Founder of "Community Ecology" and resident editor of "Ecology Document Sharing" (WeChat Official Accounts—public-facing channels on WeChat for sharing articles and educational content; ~160,000 article reads).

Honours: Chinese Bryology Annual Conference (first prize, graduate presentation); Chinese Bryophyte Congress (second prize, graduate presentation); China Agricultural University (first-class scholarship; Farming Cup first prize).

Academic translation: Joint Species Distribution Modelling: With Applications in R; Metacommunity Ecology (forthcoming).

PUBLICATIONS AND WORKS IN PROGRESS

Manuscripts and software are listed first, followed by published articles and other scholarly outputs in reverse chronological order.

WORKS IN PROGRESS & SOFTWARE

[31] **Gu, J.**, et al. *WorldBryDat: A global bryophyte functional-trait database (6,123 species; ~230 trait fields).* (Database in development; accompanying manuscript in preparation; working title).

[30] **Gu, J.**, Shao, X., Lou, A., & Ovaskainen, O. *HmscEE: Linking community ecology, lineage evolution and dynamic palaeogeography through deep time.* (Manuscript in preparation). GitHub: https://github.com/gujiqi/HmscEcoEvo

[29] **Gu, J. (contributor).** *JSDM Studio: Software for joint species distribution modelling.* (Manuscript in preparation). (Software; GitHub: https://github.com/gujiqi/JSDM_Studio).

[28] **Gu, J.**, Song, X., Ye, Y., Wang, Y., Wu, H., Shao, X., Lou, A., & Ovaskainen, O. *Environmental Filtering, Trait Differentiation, and Evolutionary Conservatism Shape Bryophyte Community Assembly on the Tibetan Plateau. Journal of Ecology* (under review).

[27] **Gu, J.**, Shao, X., Chang, Y., Roslin, T., & Ovaskainen, O. *Urbanization may alter the pathway of community change, rather than simply its rate.* (LIFEPLAN study; manuscript in preparation).

2026

[26] Zhang, Z., Bede-Fazekas, Á., García Molinos, J., Mammola, S., Kass, J. M., Qu, J., Oeser, J., Yuan, S., Zhang, C., **Gu, J.**, Ding, L., & Lin, Q. (2026). Integrating expert range maps and opportunistic occurrence records of marine fish species in range estimates. Conservation Biology, 40(2), e70154. https://doi.org/10.1111/cobi.70154

[25] Liu, Y., Liu, S., Bede-Fazekas, Á., Mammola, S., Ding, L., **Gu, J.**, Nakamura, G., Lin, Q., Wang, D., & Zhang, Z. (2026). Factors influencing species distribution model performance in tropical reef fishes. Diversity and Distributions, 32(2), e70160. https://doi.org/10.1111/ddi.70160

[24] **Gu, J.**, Lai, J., Wang, Y., Wu, H., Zhang, X., Song, X., Shao, X., & Lou, A. (2026). Theoretical foundations, methodological advances, and applications of joint species distribution models with a focus on the HMSC framework in ecology. Biodiversity Science, 34(1), 25364. https://doi.org/10.17520/biods.2025364

2025

[23] Song, X., Hong, J., **Gu, J.**, Shao, X., & Wang, R. (2025). Two newly recorded species of Anthocerotophyta from Xizang, China. Xibei Zhiwu Xuebao, 45(1), 158–162.

[22] Zhang, Z., Kass, J. M., Bede-Fazekas, Á., Mammola, S., Qu, J., Molinos, J. G., **Gu, J.**, ... & Lin, Q. (2025). Differences in predictions of marine species distribution models based on expert maps and opportunistic occurrences. Conservation Biology, e70015. https://doi.org/10.1111/cobi.70015

2024

[21] Tu, W., Lu, W., **Gu, J.**, & Lou, A. R. (2024). The species diversity and phylogenetic structure patterns of desert plant communities in the Turpan-Hami region, Xinjiang. Global Ecology and Conservation, 55, e03239. https://doi.org/10.1016/j.gecco.2024.e03239

[20] Huang, H., Zhang, Z., Bede-Fazekas, Á., Mammola, S., **Gu, J.**, Zhou, J., ... & Lin, Q. (2024). Cross-validation matters in species distribution models: a case study with goatfish species. Ecography, 2025(1), e07354. https://doi.org/10.1111/ecog.07354

[19] **Gu, J.**, Song, X., Ye, Y., Shao, X., Liao, Y., & Shao, X. (2024). Exploring the driving factors of bryophyte assemblage distribution patterns in Tibet. Frontiers in Ecology and Evolution, 12, 1376263. https://doi.org/10.3389/fevo.2024.1376263

[18] Zheng, G#., **Gu, J.**#, Zhao, W., Zhang, Y., Guan, Z., Lei, M., & He, C. (2024). Spatial, Geographical, Climatic, and Edaphic Influences on Moss Community Structure: A Case Study from Qinhuangdao, China. Forests, 15(3), 424. (#These authors contributed equally as co-corresponding authors) https://doi.org/10.3390/f15030424

[17] **Gu, J.**, Chen, J., & Lai, J. (2024). Application of large language models in biodiversity research. Biodiversity Science, 24258. https://doi.org/10.17520/biods.2024258

[16] Song, X., **Gu, J.**, Ye, Y., Wang, M., Wang, R., Ma, H., & Shao, X. (2024). Exploring Intraspecific Trait Variation in a Xerophytic Moss Species *Indusiella thianschanica* (Ptychomitriaceae) across Environmental Gradients on the Tibetan Plateau. Plants, 13(7), 921. https://doi.org/10.3390/plants13070921

[15] Song, X., **Gu, J.**, Liu, L., Liao, Y., Ma, H., Wang, R., ... & Shao, X. (2024). Exploring the distribution and habitat preferences of Polytrichaceae (Bryophyta) in Tibet, China. Heliyon, 10(14). https://doi.org/10.1016/j.heliyon.2024.e34515

[14] Sun, Y., Song, X., Chen, C., Li, S., **Gu, J.**, & Shao, X. (2024). A New Record of Pogonatum tahitense (Polytrichaceae) from Tibet, China: Taxonomic Description, Range Expansion, and Biogeographic History. Plants, 13(6), 846. https://doi.org/10.3390/plants13060846

2023

[13] Song, X.†, **Gu, J.**†, Ye, Y., Li, W., Liao, Y., Wang, R., Ma, H., & Shao, X. (2023). The Diversity and Community Pattern of Liverworts on Sygera Mountain, Tibet. Forests, 15(1), 48. (†These authors contributed equally to this work) https://doi.org/10.3390/f15010048

[12] Liao, Y., Song, X., Ye, Y., **Gu, J.**, Wang, R., Z., Zhao, D., & Shao, X. (2023). Climate Change May Pose Additional Threats to the Endangered Endemic Species Encalypta buxbaumioidea in China. Diversity, 15(2), 269. https://doi.org/10.3390/d15020269

[11] Hong, J., Song, X., **Gu, J.**, Li, W., Dong, Y., Zhuo, G., & Shao, X. (2023). Calyculariaceae, a newly recorded family of bryophytes in Tibet, China. Journal of Plant Resources and Environment, 32(5), 95–97. Journal full text (PDF)

2022

[10] **Gu, J.**, Song, X., Liao, Y., Ye, Y., Wang, R., Ma, H., & Shao, X. (2022). Tree Species Drive the Diversity of Epiphytic Bryophytes in the Alpine Forest Ecosystem: A Case Study in Tibet. Forests, 13(12), 2154. https://doi.org/10.3390/f13122154

[9] **Gu, J.**, Song, X., Liao, Y., Ye, Y., Wang, R., Ma, H., & Shao, X. (2022). Elevation, climate, and tree property drive the diversity of epiphytic bryophytes in the alpine forest ecosystem, Tibet. The International Association for Ecology Conference, Geneva, Switzerland (Poster presented at the International Association for Ecology Conference).

[8] **Gu, J.** (2022). Application of Augmented Reality in the Field of Water Environment. Liu, B., Xie, G., Cao, G., Zheng, G. (eds.), Water Environment Assessment and Intelligent Prediction in Typical Areas of Songliao Basin. Beijing: Science Press.

2021

[7] **Gu, J.**, Song, X., Liao, Y., Jiang, X., & Shao, X. (2021). Species diversity and spatial distribution pattern of epiphytic bryophytes in alpine forest ecosystem of Sejila Mountain. The 9th East Asian Federation of Ecological Societies International Congress, Hohhot, China (Oral Report and Conference Abstract).

[5] **Gu, J.**, Tong, Z., Wei, Y., Liu, X., Feng, W., Zhu, X., & Zhang, S. (2021). Dynamics of Soil Available Nitrogen Influenced by Freeze-Thaw Cycling in the Profiles of Plough Layer in Mollisol. Journal of Soil and Water Conservation, 35(2), 295-302. https://doi.org/10.13870/j.cnki.stbcxb.2021.02.039

2016–2020

[6] **Gu, J.**, Liu, L., Liao, Y., Song, P., & Shao, X. (2020). Application of Neural Network in Bryophyte Ecology. The 1st Chinese Bryophyte Congress (pp. 9). (Oral Report and Conference Abstract)

[4] **Gu, J.** Application of Artificial Intelligence Technology. (2020). In: Gao, F., Liu, W., Zheng, G. (eds.), Water Environment Monitoring and Evaluation of Provincial Buffer Zone in Songhua River Basin. Beijing: Science Press.

[3] Liu, B., Ma, X., Zheng, G., Cao, G., **Gu, J.**, Yu, J., & Xie, G. (2020). Water Quality Assessment System for Typical Rivers and Lakes of the Songhua River Using Augmented Reality (AR) Technology (Computer Software Copyright No. 2020SR1185814).

[2] Zhang, D., Zheng, G., Zhao, W., Jia, X., Zhao, Y., & **Gu, J.** (2018). Ecological Risk Monitoring System for the Nenjiang River Basin Based on Mobile App (Computer Software Copyright No. 2018SR108640).

[1] Zheng, G., Wang, Z., He, J., & **Gu, J.** (2017). Water quality assessment of the Songhua River provincial boundary buffer zone based on matter-element analysis [J]. Northeast Water Resources & Hydropower, 35(12), 25-27. https://doi.org/10.14124/j.cnki.dbslsd22-1097.2017.12.010